

AFZAL KHAN

AI Engineer | Applied AI, LLM Agents & Generative AI

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PROFESSIONAL SUMMARY

AI Engineer who ships production-grade AI systems end to end: autonomous LLM agents, generative image and video pipelines, retrieval-augmented generation (RAG), and AI security testing. First technical hire at Tamada Media, now owning AI development across Dridha Technologies' consumer products, from model behaviour and prompt design to backend, infrastructure, and UX trust. Systems-first and reliability-focused, with 6+ shipped production AI systems and 4 substantial open-source projects.

CORE SKILLS

Generative AI & Media: Text, image and video generation · FLUX (fal.ai) · ComfyUI and diffusion workflows · Google Gemini (Flash, Pro, Vision) · Nano Banana · AI image editing · image restoration and upscaling (CodeFormer, Real-ESRGAN) · virtual try-on · Sharp

LLM Systems & Agents: Prompt engineering · few-shot and style-example tuning · tool and function calling · agent planning · structured JSON output · guardrails · self-correction loops · human-in-the-loop approval · RAG · vector search

Agentic Automation & Scraping: Autonomous browser agents (Playwright) · multi-source scraping · dynamic DOM handling · CMS publishing · scheduled background tasks

AI Security: OWASP LLM Top 10 · LLM red-teaming · prompt-injection and jailbreak testing · PII and secret detection

Model Training & Fine-Tuning: LoRA fine-tuning · dataset curation · model evaluation and alignment · RLHF-style human-in-the-loop feedback

Backend & Infrastructure: Python · FastAPI · Flask · Node.js · Supabase (Postgres, Edge Functions, pg_cron) · Prisma · REST APIs · async and concurrency · queues · Docker · AWS SES · Railway · Vercel

Full-Stack & Mobile: Next.js · React · TypeScript · React Native / Expo · Tailwind CSS · Zustand · TanStack Query

Media, CV & Platforms: Image analysis · pose and identity preservation · WhisperX alignment · FFmpeg · OpenAI (GPT-4o) · Anthropic Claude · Hugging Face · Groq · Ollama

PROFESSIONAL EXPERIENCE

Tamada Media / Dridha Technologies — Associate AI Engineer

2025 – Present

- First technical hire; own AI development across Dridha Technologies (Tamada Media's AI product startup) covering autonomous agents, generative media, and full-stack AI products
- Built CUTROOM, an autonomous AI video editor, solo: WhisperX word-level alignment (± 50 ms), Gemini vision for per-clip quality checks, LLM-planned edit decision lists, FFmpeg rendering, and a self-correcting review loop that critiques and re-renders its own output
- Engineered Kiwikoo's virtual try-on engine (FLUX + Gemini): garment preprocessing, pose and identity anchoring, perceptual polish, a RAG-driven prompt system, and a concurrency-limited generation queue for production scale
- Built an autonomous AI news editor (Playwright + GPT-4o / Gemini) that scrapes 8+ trusted sources, translates to Telugu, runs image-quality checks, and publishes to a CMS end to end, with orchestration, validation, memory, and a full test suite
- Shipped an image toolchain: FLUX generation plus a CodeFormer + Real-ESRGAN restoration and upscaling service, packaged as a Dockerized FastAPI microservice deployed on Railway
- Built the Shortly email agent on Supabase (Edge Functions + pg_cron): RSS scrape, GPT-4o summarisation, human QA, and Amazon SES delivery with category-specific editorial agents; also delivered LLM-driven dashboards, browser automation, influencer discovery, and a Telugu AI voice generator

Outlier — Freelance AI Contributor

Mar 2024 – Feb 2025

- Trained and evaluated image-text models with a focus on alignment, data quality, and annotation accuracy
- Reduced model failure modes and improved robustness through structured human-in-the-loop feedback

AIMER Society (AIMERS) — AI/ML & Cybersecurity Intern

May 2023 – Jan 2024

- Built WikiWizard, a multilingual Wikipedia summarisation chatbot with memory and voice input/output; won the internship contest
- Developed multiple internal AI tools and fast summarisation and retrieval pipelines

INDEPENDENT PROJECTS

Sentinel AI — AI Security Testing Platform FastAPI · React · Python [Live Demo](#) · [GitHub](#)

- Built an open-source platform that security-tests LLMs, GitHub repositories, and live websites, returning a 0–100 security score with OWASP LLM Top 10 mapping and concrete remediation guidance
- Shipped 51 built-in attacks across 8 categories, multi-provider support (OpenAI, Anthropic, Gemini, Groq, Ollama, Hugging Face, custom endpoints), and an autonomous red-team agent that adapts jailbreaks round by round
- Engineered a heuristic evaluator (PII, secret, and refusal detection) and an async, concurrency-limited test runner with rate-limit backoff

Nori — AI-Native Productivity Browser Electron · React · TypeScript [GitHub](#)

- Built a desktop browser with an agentic AI sidebar that acts on live pages: tool calling (search, navigate, extract to CSV, compose, PDF) behind a human-approval gate
- Implemented local embeddings for semantic memory over browsing history (on-device RAG) and scheduled background research missions; approximately 10,000 lines of code

Stride — Daily Momentum Companion (Mobile) React Native · Expo · Reanimated [GitHub](#)

- Built an on-device momentum-scoring model (3-pillar, exponentially smoothed) and an adaptive UI driven by real-time solar math, with sensor-based step and sleep tracking

PikoOS — Browser-Based Desktop Environment Vanilla JS · IndexedDB [Live Demo](#) · [GitHub](#)

- Built a from-scratch, local-first web desktop (window manager, file manager, terminal, media apps) with no framework or backend

EDUCATION

B.Tech, Computer Science Engineering (Cyber Security) — GIET College of Engineering, Rajahmundry

Jul 2021 – Jul 2025

Intermediate (Class XII) — Tirumala Junior College, Visakhapatnam

2019 – 2021